

1 Candidate signals

- Oil rents ($t-1$)
- Regulatory Quality estimate (WGI 2025 Revision; standard-unit scale, $t-1$)
- Trade excluded: wholly missing for Nigeria and Trinidad and Tobago

2 Construct provenance & coverage audit

- Annual inward FDI net flow, not inward FDI stock
- Signed asinh retains zero and 504 negative-flow years
- 31 oil-rent-intensive economies restored; Nigeria included
- RQ is a perception-based national proxy; construct validity is not established by this audit

3 Benchmark competitiveness

- Oil-rents test: M2 versus strongest naïve benchmark
- RQ test: M3 versus strongest naïve benchmark
- Benchmarks: zero flow, training-sample mean, lagged flow
- Question: is the surrounding model useful out of sample?

4 Matched signal increment

- Oil rents: $\Delta R^2 = R^2(M2) - R^2(M1)$
- Regulatory Quality: $\Delta R^2 = R^2(M3) - R^2(M2)$
- Adjacent rungs evaluated on identical country-year observations
- Question: what does the signal add beyond the preceding model rung?

5 Validation & uncertainty

- 3 algorithms $\times$ 6 validation environments = 18 units/setup
- 1 pooled unseen-country unit + 5 future cutoffs
- 2,000-replicate paired country-cluster bootstrap
- Final grid fixed before verdict generation and interpretation

6 Admissibility decision

- Five unit verdicts: material, marginal, setup-dependent, unsupported, coverage-limited
- 10 setups per signal aggregated by locked cross-setup rules
- Output: candidate pre-weighting role; no AI-MCDM weights assigned
- Domain review and preference elicitation follow only after admission

Audit principles

- 1 Benchmark competitiveness is necessary but insufficient for signal admissibility.
- 2 Admissibility belongs to a signal within a defensible setup; it is not intrinsic to the variable.
- 3 Weighting follows empirical admission and domain review; weighting cannot repair a weak signal.